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November 9, 2026

The Editor
The Annals of Applied Statistics

Dear Editor,

We submit the manuscript "Escalation or shared episodes? A joint spatio-temporal model of armed conflict and social unrest", by Stefano Castruccio and Richard Marcantonio, for consideration in *The Annals of Applied Statistics*.

Protests and riots are often followed by armed conflict in the same place, and this association is widely read as escalation and used to forecast conflict. It can also arise because both kinds of events respond to the same latent episodes, such as a local political crisis or a period of media attention, with no effect of one on the other. The distinction matters for the study of political violence and for early warning, and it is a statistical question: we show that models of conflict alone, including models with their own latent space-time field, cannot make it, and that a joint model of the two outcomes can.

The paper proposes a joint latent Gaussian model for the weekly occurrence of conflict and unrest on a 0.5° grid, in which the two outcomes share a space-time field with an estimated loading, alongside cross-lagged effects of their recent histories. The fields are discretised by finite volumes directly on the sphere, with sea barriers, and fitted by an empirical Bayes Laplace approximation; case-control subsampling of non-events makes the hyperparameter estimation feasible on millions of cell-weeks per outcome. Simulation studies show that univariate models report escalation that does not exist, while the joint model has close to nominal coverage. In ACLED data for West Africa and the Sahel and for Africa and the Middle East, 2018–2024, shared episodes account for between a quarter and two thirds of the association between recent unrest and conflict. We also examine what the joint model adds to prediction, and find that its value lies in filling in an outcome where it goes unrecorded while the other is observed, rather than in forecasting when both are observed.

We believe the paper suits the journal because the substantive question drives the statistical development, and because the methods, a multivariate latent Gaussian model on the sphere with scalable inference for rare binary outcomes, apply to other multivariate event data in the social and environmental sciences. The Supplementary Material contains the construction of the data, computational details, simulation studies of the univariate building blocks, and further results. Code for all analyses will be made available; the ACLED data are available from ACLED under its terms of use.

The manuscript is not under consideration elsewhere, and both authors have approved the submission. The use of a generative AI assistant in preparing the code and the text is described in the Acknowledgments. We

have no competing interests to declare.

Sincerely,

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